

Charles Lepine

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U.S. Citizen - eligible to obtain U.S. Government security clearance

EDUCATION

North Carolina State University | Raleigh, NC

B.S. Computer Engineering | Senior Standing | Aug 2022 - Dec 2026

Relevant Coursework: Fundamentals of Logic Design, Microelectronics, Electric Circuits, Introduction to Embedded Systems, Circuit Board Layout, Linear Systems, Computer Networking, Data Structures & OOP (C++), Computer Systems Programming

In Progress (Fall 2026): Design of Complex Digital Systems (Verilog/Vivado), Microprocessor Architecture, Communication Engineering

TECHNICAL SKILLS

Hardware & Digital Design: Verilog, PCB layout (KiCad), surface-mount soldering, H-bridge motor drive, ADCs, timers, interrupts

Test & Lab Equipment: Oscilloscopes, logic analyzers, multimeters, Analog Discovery 3, Waveforms, datasheet-driven bring-up

Languages: C, C++, Assembly, Python, Java, PowerShell

Software & Tools: Code Composer Studio, MATLAB, Wireshark, Git, GitHub, VS Code, CLion, NVIDIA DeepStream SDK

EXPERIENCE

IBM | Software Developer Intern - RTP, NC

May 2026 - Aug 2026

SSL/TLS Certificate Lifecycle Automation

- Built a PowerShell automation system for IBM HTTP Server that cut TLS certificate deployment from a 1-3 day manual process to a 5-minute setup, with every subsequent renewal then completing automatically and unattended in a few minutes.
- Designed a modular challenge-plugin architecture of 6 plugins supporting HTTP and DNS validation across AWS, Azure, Cloudflare, and Google Cloud, making the tool compatible with any ACME-compliant certificate authority on the market.
- Consolidated 20 manual renewal steps into a single script with encrypted credential resolution from cloud vaults, automatic rollback on failed deployment, and email alerting on every renewal event; validated across 13 separate production domains.

AI Agent Extraction Pipeline

- Designed a three-stage pipeline (Slack ingestion, preprocessing, and LLM extraction) that applies PII and secrets guardrails and turns raw engineering discussion threads into a structured, searchable FAQ page in under 10 minutes of runtime.
- Orchestrated three sequential LLM calls (question extraction, answer extraction, and topic classification) in LangChain via LiteLLM, enforcing consistent, machine-parsable structured output with a Pydantic schema at every stage of the chain.
- Extended the pipeline to additional source types with an adaptive segmentation algorithm that reused the LLM extraction stage unmodified, and added a checkpoint system so that repeat runs skip any threads that were already processed.

NCFC / Wake FC / Net Sports | Soccer Referee - Raleigh-Durham, NC

June 2020 - Present

- Officiated over 500 matches from amateur leagues through semi-professional UPSL play, enforcing FIFA and league laws.
- Mentored junior and assistant referees on positioning, mechanics, and rule interpretation to ensure consistency.

PROJECTS

AI Occupancy Detection Camera - NVIDIA PeopleNet, DeepStream SDK

Senior Design | Aug 2026 - Present

- Leading a team of 4 building an AI occupancy camera that counts people in a room and streams live headcount to a dashboard.
- Architecting the vision pipeline around NVIDIA PeopleNet for person detection, deployed through the NVIDIA DeepStream SDK.
- Owning scope breakdown and integration across the team, from camera capture through inference to the dashboard front end.

Remote-Operated Embedded Vehicle - C, Assembly, MSP430

Intro to Embedded Systems

- Soldered custom PCBs with surface-mount components for the power, display, and control subsystems of the vehicle.
- Programmed an MSP430 in C and assembly to drive H-bridge motors, sample sensors over ADC, and service timer interrupts.
- Integrated a Wi-Fi module and infrared line-detection, verifying signal behavior on an oscilloscope and logic analyzer.

Digital Logic Design Project - Verilog, CAD tools, Analog Discovery 3

Fundamentals of Logic Design

- Designed and simulated digital logic systems in Verilog, using CAD tools for synthesis, minimization, and state assignment.
- Assembled and wired a combinational logic circuit on a breadboard, integrating ICs with input and output components.
- Verified logic behavior against expected waveforms with the Analog Discovery 3 logic analyzer and pattern generator.

LEADERSHIP & ACTIVITIES

NC State Club Soccer - Team Member | Jan 2024 - Present

- Selected from 200+ candidates in a competitive 3-day tryout.

NC State Rocket League Esports - Team Member | Aug 2022 - Present

- Top 0.01% global ranking, \$5,000+ tournament wins.